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Inventor(s)	Rowe; Michael P.

Storage compartment assemblies having artificial muscles and methods of operating storage compartment assemblies

Abstract

A storage compartment assembly including a storage compartment having at least one inside surface and defining an opening for receiving an object, one or more artificial muscles disposed within the storage compartment along the inside surface. Each artificial muscle includes a housing having an electrode region and an expandable fluid region, a dielectric fluid housed within the housing, and an electrode pair positioned in the electrode region of the housing. The electrode pair includes a first electrode and a second electrode. The electrode pair is configured to actuate between a non-actuated position and an actuated position such that actuation from the non-actuated position to the actuated position directs the dielectric fluid into the expandable fluid region to expand the expandable fluid region into the storage compartment.

Inventors: **Rowe; Michael P. (Pinckney, MI)**

Applicant: **Toyota Motor Engineering & Manufacturing North America, Inc.** (Plano, TX)

Family ID: **85798281**

Assignee: **TOYOTA MOTOR ENGINEERING & MANUFACTURING NORTH AMERICA, INC. (Pl.ano, TX)**

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Primary Examiner: Cheung; Chun Hoi

Attorney, Agent or Firm: Dinsmore & Shohl LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims the benefit of U.S. Patent Application No. 63/253,361, filed Oct. 7, 2021, for “Compartment Comprising Artificial Muscles,” which is hereby incorporated by reference in its entirety including the drawings.

TECHNICAL FIELD

(1) The present specification generally relates to storage compartment assemblies and methods for operating storage compartment assemblies, and, more specifically, storage compartment assemblies having artificial muscles and methods for operating storage compartment assemblies having artificial muscles.

BACKGROUND

(2) Traditional storage compartments in vehicles are capable of storing small objects, such as sunglasses, keys, and the like. However, these small objects can move around in the storage compartment, causing noise and damage to the object. Additionally, movement of the vehicle may result in the objects being unintentionally removed from the storage compartment. Accordingly, a need exists for storage compartment assemblies with mechanisms for retaining small objects while a vehicle is in motion.

SUMMARY

(3) In one embodiment, a storage compartment assembly includes a storage compartment having at least one inside surface and defining an opening for receiving an object, one or more artificial muscles disposed within the storage compartment along the inside surface. Each artificial muscle includes a housing having an electrode region and an expandable fluid region, a dielectric fluid housed within the housing, and an electrode pair positioned in the electrode region of the housing. The electrode pair includes a first electrode and a second electrode. The electrode pair is configured to actuate between a non-actuated position and an actuated position such that actuation from the non-actuated position to the actuated position directs the dielectric fluid into the expandable fluid region to expand the expandable fluid region into the storage compartment.

(4) In another embodiment, a storage compartment assembly includes a storage compartment having at least one inside surface and defining an opening for receiving an object, and a compartment portal extending over the opening. The compartment portal includes a plurality of artificial muscle cantilevers. Each artificial muscle cantilever includes a housing with an electrode region fluidly coupled to a cantilevered region, the cantilevered region having at least one expandable fluid chamber, a first electrode and a second electrode each disposed in the electrode region of the housing, and a bladder housing a fluid and extending into the cantilevered region. The cantilevered region of the housing of each of the plurality of artificial muscle cantilevers extends over the opening.

(5) In yet another embodiment, a method of operating a storage compartment assembly, the method includes actuating a plurality of artificial muscle cantilevers of a compartment portal, and actuating one or more artificial muscles disposed within the storage compartment. Each artificial muscle cantilever includes a housing with an electrode region fluidly coupled to a cantilevered region, the cantilevered region having at least one expandable fluid chamber, a first electrode and a second electrode each disposed in the electrode region of the housing, and a bladder housing a fluid and extending into the cantilevered region. The cantilevered region of the housing of each of the plurality of artificial muscle cantilevers extends over an opening for receiving an object, the opening defined by a storage compartment. The first electrode and the second electrode of the plurality of artificial muscle cantilevers are actuatable between a non-actuated position and an actuated position such that actuation from the non-actuated position to the actuated position directs the fluid in the bladder into the at least one expandable fluid chamber of the cantilevered region. Each artificial muscle includes a housing with an electrode region and an expandable fluid region, a dielectric fluid housed within the housing, and an electrode pair positioned in the electrode region

of the housing. The electrode pair includes a first electrode and a second electrode. The electrode pair is configured to actuate between a non-actuated position and an actuated position such that actuation from the non-actuated position to the actuated position directs the dielectric fluid into the expandable fluid region.
(6) These and additional features provided by the embodiments described herein will be more fully understood in view of the following detailed description, in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The embodiments set forth in the drawings are illustrative and exemplary in nature and not intended to limit the subject matter defined by the claims. The following detailed description of the illustrative embodiments can be understood when read in conjunction with the following drawings, where like structure is indicated with like reference numerals and in which:
(2) FIG. **1** schematically depicts a top view of a storage compartment assembly, according to one or more embodiments shown and described herein;
(3) FIG. **2** schematically depicts a cross-sectional side view of the storage compartment assembly of FIG. **1** taken along lines A-A with the storage compartment assembly in a non-actuated state, according to one or more embodiments shown and described herein;
(4) FIG. **3** schematically depicts a cross-sectional side view of the storage compartment assembly of FIG. **1** taken along lines A-A with the storage compartment assembly in an actuated state, according to one or more embodiments shown and described herein;
(5) FIG. **4** schematically depicts a cross-sectional side view of an artificial muscle of the storage compartment assembly of FIG. **1** with the artificial muscle in a non-actuated position, according to one or more embodiments shown and described herein;
(6) FIG. **5** schematically depicts a cross-sectional side view of the artificial muscle of FIG. **4** with the artificial muscle in an actuated position, according to one or more embodiments shown and described herein;
(7) FIG. **6** schematically depicts a cross-sectional side view of an artificial muscle cantilever of the storage compartment assembly of FIG. **1** with the artificial muscle cantilever in a non-actuated position, according to one or more embodiments shown and described herein;
(8) FIG. **7** schematically depicts a cross-sectional side view of the artificial muscle cantilever of FIG. **6** with the artificial muscle cantilever in an actuated position, according to one or more embodiments shown and described herein; and
(9) FIG. **8** schematically depicts a control system for operating the storage compartment assembly of FIG. **1**, according to one or more embodiments shown and described herein.
DETAILED DESCRIPTION
(10) Embodiments described herein are directed to storage compartment assemblies including a storage compartment, a plurality of artificial muscles positioned within the storage compartment, and a compartment portal positioned across an opening of the storage compartment. The artificial muscles are configured to expand within the storage compartment to contact and retain an object positioned within the storage compartment. The compartment portal includes a plurality of artificial muscle cantilevers that extend across the opening. The artificial muscle cantilevers may be actuated to stiffen the artificial muscle cantilevers, thereby restricting egress and ingress of an object within the storage compartment.
(11) Referring now to FIG. **1**, a storage compartment assembly **10** is depicted. The storage compartment assembly **10** includes a storage compartment **12**, a plurality of artificial muscles **14** (FIG. **2**) disposed within the storage compartment **12**, and a compartment portal **16**. The storage compartment **12** may be formed in a surface **1** of a vehicle, such that the storage compartment **12**

may extend into one of a dashboard, an overhead console, or a panel of the vehicle. In embodiments, the storage compartment **12** may be formed in a surface external to a vehicle, such as a building or other structure. As depicted in FIG. **1**, the opening **24** may be shaped as an oval. However, the opening **24** may have any shape operable to act as an opening for the storage compartment **12**, such as square, rectangular, circular, or the like.
(12) Referring to FIGS. **2** and **3**, the storage compartment **12** may include a plurality of inside surfaces, including an end surface **20** and one or more sidewalls **22** extending from the end surface **20**, and an opening **24** defined by upper portions **26** of the sidewalls **22** of the storage compartment **12**. The artificial muscles **14** may be disposed within the storage compartment **12** along the sidewalls **22** and the end surface **20**. The compartment portal **16** may extend across the opening **24** of the storage compartment **12**.
(13) The artificial muscles **14** may be actuated between a non-actuated position and an actuated position. When moving from the non-actuated position to the actuated position, the artificial muscles **14** may expand toward a center **28** of the storage compartment **12**. The center **28** of the storage compartment **12** may be a central portion of the storage compartment **12** in a vertical direction (e.g., in the +/−Z direction), a longitudinal direction (e.g., in the +/−Y direction), a lateral direction (e.g., in the +/−X direction), or any combination thereof.
(14) The compartment portal **16** may include a plurality of artificial muscle cantilevers **30** that extend across the opening **24** of the storage compartment **12**. The artificial muscle cantilevers **30** may be actuated between an actuated position and a non-actuated position. In the actuated position, a stiffness of the artificial muscle cantilevers **30** is increased, as compared to a stiffness of the artificial muscle cantilevers **30** when in the non-actuated position, to prevent egress and ingress of objects into and out of the storage compartment **12** through the opening **24**. Accordingly, when in the non-actuated position, the stiffness of the artificial muscle cantilevers **30** decreases such that the artificial muscle cantilevers **30** may be deformable and flexible to allow access to the storage compartment **12** through the opening **24**.
(15) The storage compartment **12** may be configured to receive an object **2** (FIG. **3**). The storage compartment assembly **10** may be configured to retain the object **2** within the storage compartment **12**. The storage compartment assembly **10** may be movable between a non-actuated state (FIG. **2**) and an actuated state (FIG. **3**).
(16) Referring to FIG. **2**, in the non-actuated state, the artificial muscles **14** are in the non-actuated position, and the artificial muscle cantilevers **30** are in the non-actuated position. In the non-actuated state, the object **2** is permitted to pass through the compartment portal **16** to be inserted into or removed from the storage compartment **12**.
(17) Referring to FIG. **3**, in the actuated state, the artificial muscles **14** are in the actuated position, and the artificial muscle cantilevers **30** are in the actuated position. In the actuated state, the artificial muscle cantilevers **30** may restrict the object **2** from passing through the compartment portal **16**. In the actuated state, the artificial muscles **14** may expand to contact and deform around the object **2** positioned within the storage compartment **12**. The contact between the artificial muscles **14** and the object **2** maintains the position of the object **2** within the storage compartment **12**.
(18) Referring now to FIGS. **4** and **5**, an embodiment of the artificial muscle **14** is depicted in more detail. Specifically, the artificial muscle **14** may include a housing **102**, an electrode pair **104**, including a first electrode **106** and a second electrode **108**, fixed to opposite surfaces of the housing **102**, a first electrical insulator layer **110** fixed to the first electrode **106**, and a second electrical insulator layer **112** fixed to the second electrode **108**. In some embodiments, the artificial muscle **14** may include only a single electrical insulator layer fixed to one of the first electrode **106** and the second electrode **108** leaving the other of the first electrode **106** and the second electrode **108** non-electrically insulated. In some embodiments, the housing **102** is a one-piece monolithic layer including a pair of opposite inner surfaces, such as a first inner surface **114** and a second inner

surface **116**, and a pair of opposite outer surfaces, such as a first outer surface **118** and a second outer surface **120**. In some embodiments, the first inner surface **114** and the second inner surface **116** of the housing **102** are heat-sealable. In other embodiments, the housing **102** may be a pair of individually fabricated film layers, such as a first film layer **122** and a second film layer **124**. Thus, the first film layer **122** includes the first inner surface **114** and the first outer surface **118**, and the second film layer **124** includes the second inner surface **116** and the second outer surface **120**.

(19) Throughout the ensuing description, reference may be made to the housing **102** including the first film layer **122** and the second film layer **124**, as opposed to the one-piece housing. It should be understood that either arrangement is contemplated. In some embodiments, the first film layer **122** and the second film layer **124** generally include the same structure and composition. For example, in some embodiments, the first film layer **122** and the second film layer **124** each comprises biaxially oriented polypropylene (BOPP).

(20) The first electrode **106** and the second electrode **108** are each positioned between the first film layer **122** and the second film layer **124**. In some embodiments, the first electrode **106** and the second electrode **108** are each aluminum-coated polyester such as, for example, Mylar®. In addition, one of the first electrode **106** and the second electrode **108** is a negatively charged electrode and the other of the first electrode **106** and the second electrode **108** is a positively charged electrode. For purposes discussed herein, either electrode **106**, **108** may be positively charged so long as the other electrode **106**, **108** of the artificial muscle **14** is negatively charged. In embodiments in which only one of the first electrode **106** and the second electrode **108** is insulated by an electrically insulated layer, such as the first electrical insulator layer **110** or the second electrical insulator layer **112**, the insulated layer is the positively charged electrode and the non-insulated layer is the negatively charged electrode.

(21) The first electrode **106** has a film-facing surface **126** and an opposite inner surface **128**. The first electrode **106** is positioned against the first film layer **122**, specifically, the first inner surface **114** of the first film layer **122**. In addition, the first electrode **106** includes a first terminal extending from the first electrode **106** past an edge of the first film layer **122** such that the first terminal can be connected to a power supply to actuate the first electrode **106**. Specifically, the first terminal is coupled, either directly or in series, to a power supply and a controller of an control system **300**. Similarly, the second electrode **108** has a film-facing surface **148** and an opposite inner surface **130**. The second electrode **108** is positioned against the second film layer **124**, specifically, the second inner surface **116** of the second film layer **124**. The second electrode **108** includes a second terminal extending from the second electrode **108** past an edge of the second film layer **124** such that the second terminal can be connected to a power supply and a controller of the control system **300** to actuate the second electrode **108**.

(22) With respect now to the first electrode **106**, in embodiments, the first electrode **106** includes two or more fan portions **132** extending radially from a center axis C of the artificial muscle **14**. In some embodiments, the first electrode **106** includes only two fan portions **132** positioned on opposite sides or ends of the first electrode **106**. In some embodiments, the first electrode **106** includes more than two fan portions **132**, such as three, four, or five fan portions **132**. In embodiments in which the first electrode **106** includes an even number of fan portions **132**, the fan portions **132** may be arranged in two or more pairs of fan portions **132**.

(23) The second electrode **108** may similarly include two or more fan portions **154** extending radially from a center axis C of the artificial muscle **14**. In some embodiments, the second electrode **108** includes only two fan portions **154** positioned on opposite sides or ends of the second electrode **108**. In some embodiments, the second electrode **108** includes more than two fan portions **154**, such as three, four, or five fan portions **154**. In embodiments in which the second electrode **108** includes an even number of fan portions **154**, the fan portions **154** may be arranged in two or more pairs of fan portions **154**.

(24) In embodiments, a central opening **168** is formed within the second electrode **108** between the

fan portions **154**, and is coaxial with the center axis C. Each fan portion **154** has a fan length extending from a perimeter **142** of the central opening **168** to the second end **160** of the fan portion **154**. With reference to FIG. **3**, the first electrode **106** may be positioned between the second electrode **108** and the sidewall **22** such that the central opening **168** in the second electrode **108** is directed toward the center **28** of the storage compartment **12**. The expandable fluid region **196** may extend out of the opening **168** toward the center **28** of the storage compartment **12**.

(25) Referring back to FIGS. **4** and **5**, the first electrode **106**, the second electrode **108**, the first electrical insulator layer **110**, and the second electrical insulator layer **112** provide a barrier that prevents the first film layer **122** from sealing to the second film layer **124** forming an unsealed portion **192**. The unsealed portion **192** of the housing **102** includes an electrode region **194**, in which the electrode pair **104** is provided, and an expandable fluid region **196**, which is surrounded by the electrode region **194**. The central opening **168** of the second electrode **108** defines the expandable fluid region **196**. Although not shown, the housing **102** may be cut to conform to the geometry of the electrode pair **104** and reduce the size of the artificial muscle **14**, namely, the size of the sealed portion **190**.

(26) A dielectric fluid **198** is provided within the unsealed portion **192** and flows freely between the first electrode **106** and the second electrode **108**. A "dielectric" fluid as used herein is a medium or material that transmits electrical force without conduction and as such has low electrical conductivity. Some non-limiting example dielectric fluids include perfluoroalkanes, transformer oils, and deionized water. It should be appreciated that the dielectric fluid **198** may be injected into the unsealed portion **192** of the artificial muscle **14** using a needle or other suitable injection device.

(27) The expandable fluid region **196** of the artificial muscles **14** may be larger than that depicted in FIGS. **4** and **5**. The artificial muscles **14** may include additional dielectric fluid **198** within the housing **102** and/or the first electrode **106** and the second electrode **108** may be displaced further during actuation of the artificial muscle **14** to displace additional fluid and further expand the expandable fluid region **196**.

(28) In embodiments, the first electrode **106** may include structure similar to the second electrode **108**, with a central opening disposed opposite the central opening **168** of the second electrode **108**. In such embodiments, the dielectric fluid **198** may extend out of the central opening in the first electrode **106** and the central opening **168** in the second electrode **108**. In embodiments, the artificial muscles **14** may include an opening at an end of either of the first electrode **106** and the second electrode **108** such that the dielectric fluid extends out of a side of the artificial muscle **14**.

(29) In the non-actuated position, as shown in FIG. **4**, the first electrode **106** and the second electrode **108** are spaced apart with the dielectric fluid **198** positioned therebetween in the electrode region **194**. In the non-actuated position, the expandable fluid region **196** has a first height H**1**.

(30) In the actuated position, as shown in FIG. **5**, the first electrode **106** and the second electrode **108** are brought into contact with and oriented parallel to one another to force the dielectric fluid **198** into the expandable fluid region **196**. This causes the dielectric fluid **198** to flow through the central opening **168** of the second electrode **108** and inflate the expandable fluid region **196**.

(31) When actuated, as shown in FIG. **5**, the first electrode **106** and the second electrode **108** zipper toward one another from the second ends **136**, **158** of the fan portions **132**, **154** thereof, thereby pushing the dielectric fluid **198** into the expandable fluid region **196**. As shown, when in the actuated position, the first electrode **106** and the second electrode **108** are parallel to one another. In the actuated position, the dielectric fluid **198** flows into the expandable fluid region **196** to inflate the expandable fluid region **196**. As such, the first film layer **122** and the second film layer **124** expand in opposite directions. In the actuated position, the expandable fluid region **196** has a second height H**2**, which is greater than the first height H**1** of the expandable fluid region **196** when in the non-actuated position. Although not shown, it should be noted that the electrode pair **104** may be partially actuated to a position between the non-actuated position and the actuated position.

This would allow for partial inflation of the expandable fluid region **196** and adjustments when necessary.

(32) The first electrode **106** and the second electrode **108** may be electrically connected to a first power supply **306**. In order to move the first electrode **106** and the second electrode **108** toward one another, a voltage is applied by the first power supply **306**. In some embodiments, a voltage of up to 10 kV may be provided from the first power supply **306** to induce an electric field through the dielectric fluid **198**. The resulting attraction between the first electrode **106** and the second electrode **108** pushes the dielectric fluid **198** into the expandable fluid region **196**. Pressure from the dielectric fluid **198** within the expandable fluid region **196** causes the second film layer **124** and the second electrical insulator layer **112** to deform in a first axial direction along the center axis C of the first electrode **106**. Once the voltage being supplied to the first electrode **106** and the second electrode **108** is discontinued, the first electrode **106** and the second electrode **108** return to their initial, non-parallel position in the non-actuated position.

(33) Referring to FIGS. **6** and **7**, each artificial muscle cantilever **30** may include similar structure to that of the artificial muscles **14**. Each artificial muscle cantilever **30** may include a housing **200**, an electrode pair including a first electrode **202** and a second electrode **204**, a fluid **206** housed within the housing **200**, and a bladder **208**. The housing **200** may include a cantilevered region **210** and an electrode region **212** fluidly coupled to the cantilevered region **210**. The first electrode **202** and the second electrode **204** may be disposed within the electrode region **212** of the housing **200**. The electrode region **212** of each of the plurality of artificial muscle cantilevers **30** may be disposed at a periphery **32** of the opening **24** of the storage compartment **12**. The first electrode **202** and the second electrode **204** may include similar structure to that of the electrodes **106**, **108** of the artificial muscles **14**, and such like structure will not be described again for brevity. The first electrode **202** and the second electrode **204** may be movably attached to the housing **200** to allow the first electrode **202** and the second electrode **204** to move between the non-actuated position to the actuated position.

(34) Each of the first electrode **202** and the second electrode **204** may be electrically connected to a second power supply **314**. The second power supply **314** may send a current or voltage to the electrodes **202**, **204** to move the electrodes **202**, **204** from the non-actuated position to the actuated position.

(35) The cantilevered region **210** may have at least one expandable fluid chamber **214**. The cantilevered region **210** may include a plurality of expandable fluid chambers **214**, such as two, that are each in fluid communication with each other. The bladder **208** may be positioned within the housing **200**, and extend from the electrode region **212** into the expandable fluid chamber **214**. The bladder **208** may enclose the fluid **206**. The fluid **206** may be, for example, a dielectric fluid, an incompressible fluid, a liquid, a gas, or the like. The bladder **208** may include a compressible portion **216** positioned between the first electrode **202** and the second electrode **204**, and an expandable portion **218** extending into the expandable fluid chamber **214**. In embodiments, the artificial muscle cantilevers **30** may not include a bladder **208**, such that the electrodes **202**, **204** displace the fluid **206** throughout the expandable fluid chamber **214**. The cantilevered region **210** of the housing **200** of each of the plurality of artificial muscle cantilevers **30** may extend over the opening **24**.

(36) Referring to FIG. **6**, in the non-actuated position, the first electrode **202** and the second electrode **204** may be spaced apart, with the compressible portion **216** of the bladder **208** positioned between the first electrode **202** and the second electrode **204**. Referring to FIG. **7**, when the artificial muscle cantilevers **30** are actuated, the first electrode **202** and the second electrode **204** may be brought together to compress the compressible portion **216** of the bladder **208** to direct the fluid in the bladder **208** into the expandable portion **218**. When the compressible portion **216** is compressed, the fluid in the compressible portion **216** may extend into the expandable portion **218**. When the fluid is directed into the expandable portion **218**, the expandable portion **218** expands in

the expandable fluid chamber **214** of the cantilevered region **210**, expanding the expandable fluid chamber **214** and increasing a stiffness of the cantilevered region **210** of the housing **200**.

(37) Referring again to FIG. **1**, adjacent pairs of the plurality of artificial muscle cantilevers **30** are coupled together along their cantilevered regions **210**. The plurality of artificial muscle cantilevers **30** may include a first set **220** and a second set **222**. The first set **220** may include cantilevered regions **210** extending over a first region of the opening **24** and the second set **222** may include cantilevered regions **210** extending over a second region of the opening **24**. Ends **224** of the cantilevered regions **210** of the first set **220** of artificial muscle cantilevers **30** may be disposed to face ends **224** of the cantilevered regions **210** of the second set **222** of artificial muscle cantilevers at a location over the opening **24**. The contact between the ends **224** of the cantilevered regions **210** of the first set **220** of artificial muscle cantilevers **30** and the ends **224** of the cantilevered regions **210** of the second set **222** of artificial muscle cantilevers **30** may define a central axis C extending across the compartment portal **16**. Each of the cantilevered regions **210** of the first set **220** of artificial muscle cantilevers **30** and the cantilevered regions **210** of the second set **222** of artificial muscle cantilevers **30** may be angled obliquely from the central axis C.

(38) Referring to FIG. **8**, the storage compartment **12** may further include a control system **300**. The control system **300** may include a controller **302**, a first proximity sensor **320**, a second proximity sensor **321**, an operating device **304**, a first power supply **306**, a second power supply **314**, and a communication path **308**. The first proximity sensor **320** may be configured to detect a proximity of an object approaching the compartment portal **16**. The object, for example, may be a hand of a user. In response to receiving a signal from the first proximity sensor **320** indicative of the proximity of the object near the compartment portal **16**, the controller **302** may be configured to determine whether the detected proximity is within a predetermined proximity of the compartment portal **16** and, in response, actuate the artificial muscle cantilevers **30** to the non-actuated position. The predetermined proximity may be, for example, less than one inch, one inch, two inches, three inches, or more than three inches. The first proximity sensor **320** may be any sensor capable of detecting an object around the first proximity sensor **320**, such as, for example, a through-beam sensor, an inductive sensor, a force sensor, or the like.

(39) The second proximity sensor **321** may be positioned within the storage compartment **12**. The second proximity sensor **321** may be configured to detect the object **2** within the storage compartment **12**. In response to receiving a signal from the second proximity sensor **321** indicative of the object **2** being positioned within the storage compartment **12**, the controller **302** may be configured to actuate the artificial muscles **14** to the actuated position. When the second proximity sensor **321** does not detect the object **2** in the storage compartment **12**, the controller **302** may be configured to actuate the artificial muscles **14** to the non-actuated position. The second proximity sensor **321** may be any sensor capable of detecting an object around the second proximity sensor **321**, such as, for example, a through-beam sensor, an inductive sensor, a force sensor, or the like.

(40) The controller **302** includes a processor **310** and a non-transitory electronic memory **312** to which various components are communicatively coupled. In some embodiments, the processor **310** and the non-transitory electronic memory **312** and/or the other components are included within a single device. In other embodiments, the processor **310** and the non-transitory electronic memory **312** and/or the other components may be distributed among multiple devices that are communicatively coupled. The controller **302** includes non-transitory electronic memory **312** that stores a set of machine-readable instructions. The processor **310** executes the machine-readable instructions stored in the non-transitory electronic memory **312**. The non-transitory electronic memory **312** may comprise RAM, ROM, flash memories, hard drives, or any device capable of storing machine-readable instructions such that the machine-readable instructions can be accessed by the processor **310**. Accordingly, the control system **300** described herein may be implemented in any conventional computer programming language, as pre-programmed hardware elements, or as a combination of hardware and software components. The non-transitory electronic memory **312**

may be implemented as one memory module or a plurality of memory modules.

(41) In some embodiments, the non-transitory electronic memory **312** includes instructions for executing the functions of the control system **300**. The instructions may include instructions for operating the artificial muscles **14** and the artificial muscle cantilevers **30** based on a user command.

(42) The processor **310** may be any device capable of executing machine-readable instructions. For example, the processor **310** may be an integrated circuit, a microchip, a computer, or any other computing device. The non-transitory electronic memory **312** and the processor **310** are coupled to the communication path **308** that provides signal interconnectivity between various components and/or modules of the control system **300**. Accordingly, the communication path **308** may communicatively couple any number of processors with one another, and allow the modules coupled to the communication path **308** to operate in a distributed computing environment. Specifically, each of the modules may operate as a node that may send and/or receive data. As used herein, the term "communicatively coupled" means that coupled components are capable of exchanging data signals with one another such as, for example, electrical signals via conductive medium, electromagnetic signals via air, optical signals via optical waveguides, and the like.

(43) As schematically depicted in FIG. **8**, the communication path **308** communicatively couples the processor **310** and the non-transitory electronic memory **312** of the controller **302** with a plurality of other components of the control system **300**. For example, the control system **300** depicted in FIG. **8** includes the processor **310** and the non-transitory electronic memory **312** communicatively coupled with the operating device **304**, the first power supply **306**, the second power supply **314**, the first proximity sensor **320**, and the second proximity sensor **321**.

(44) The operating device **304** allows for a user to control operation of the artificial muscles **14** and the artificial muscle cantilevers **30**. In some embodiments, the operating device **304** may be a switch, toggle, button, or any combination of controls to provide user operation. As a non-limiting example, a user may actuate the artificial muscles **14** and the artificial muscle cantilevers **30** into the actuated state by activating controls of the operating device **304** to a first position. While in the first position, the artificial muscles **14** and the artificial muscle cantilevers **30** will remain in the actuated state. The user may switch the artificial muscles **14** and the artificial muscle cantilevers **30** into the non-actuated state by operating the controls of the operating device **304** out of the first position and into a second position. The operating device **304** may individually operate the artificial muscles **14** and the artificial muscle cantilevers **30** via the first power supply **306** and the second power supply **314**, respectively.

(45) The operating device **304** is coupled to the communication path **308** such that the communication path **308** communicatively couples the operating device **304** to other modules of the control system **300**. The operating device **304** may provide a user interface for receiving user instructions as to a specific operating configuration of the artificial muscles **14** and the artificial muscle cantilevers **30**. In addition, user instructions may include instructions to operate the artificial muscles **14** and the artificial muscle cantilevers **30** only at certain conditions.

(46) The first proximity sensor **320** and the second proximity sensor **321** may be communicatively coupled to the controller **302** over the communication path **308**. The first proximity sensor **320** may send signals to the controller **302** indicative of a proximity of an object to the compartment portal **16**. The second proximity sensor **321** may send signals to the controller **302** indicative of the object **2** being positioned in the storage compartment **12**. The controller **302** may be configured to control actuation of the artificial muscles **14** and/or artificial muscle cantilevers based on the signals received from the first proximity sensor **320** and/or the second proximity sensor **321**. When the controller **302** receives a signal from the first proximity sensor **320** indicative of an object in close proximity to the compartment portal **16**, the controller **302** may move the artificial muscle cantilevers **30** from the actuated position to the non-actuated position, and may move the artificial muscles **14** from the actuated position to the non-actuated position. When the controller **302**

receives a signal from the second proximity sensor **321** indicative of the object **2** being positioned in the storage compartment **12**, the controller **302** may move the artificial muscles **14** from the non-actuated position to the actuated position.

(47) The first power supply **306** (e.g., battery) provides power to the artificial muscles **14**. The second power supply **314** provides power to the artificial muscle cantilevers **30**. In some embodiments, the first power supply **306** and the second power supply **314** are rechargeable direct current power sources. It is to be understood that the first power supply **306** and the second power supply **314** may be a single power supply or battery for providing power to the artificial muscles **14** and the artificial muscle cantilevers **30**. A power adapter (not shown) may be provided and electrically coupled via a wiring harness or the like for providing power to the artificial muscles **14** and the artificial muscle cantilevers **30** via the first power supply **306** and the second power supply **314**. By providing power to the artificial muscles **14** and the artificial muscle cantilevers **30**, the first power supply **306** and the second power supply **314** actuate the artificial muscles **14** and the artificial muscle cantilevers **30** to move from the non-actuated position to the actuated position.

(48) In some embodiments, the control system **300** also includes a display device **316**. The display device **316** is coupled to the communication path **308** such that the communication path **308** communicatively couples the display device **316** to other modules of the control system **300**. The display device **316** may output a notification in response to an actuation state of the artificial muscles **14** and the artificial muscle cantilevers **30** or indication of a change in the actuation state of the artificial muscles **14** and the artificial muscle cantilevers **30**. Moreover, the display device **316** may be a touchscreen that, in addition to providing optical information, detects the presence and location of a tactile input upon a surface of or adjacent to the display device **316**. Accordingly, the display device **316** may include the operating device **304** and receive mechanical input directly upon the optical output provided by the display device **316**.

(49) In some embodiments, the control system **300** includes network interface hardware **318** for communicatively coupling the control system **300** to a portable device **324** via a network **322**. The portable device **324** may include, without limitation, a smartphone, a tablet, a personal media player, or any other electric device that includes wireless communication functionality. It is to be appreciated that, when provided, the portable device **324** may serve to provide user commands to the controller **302**, instead of the operating device **304**. As such, a user may be able to control or set a program for controlling the artificial muscles **14** and the artificial muscle cantilevers **30** without utilizing the controls of the operating device **304**. Thus, the artificial muscles **14** and the artificial muscle cantilevers **30** may be controlled remotely via the portable device **324** wirelessly communicating with the controller **302** via the network **322**. The portable device **324** may receive and display the determined electrical resistance or muscle displacement from the controller **302** via the network **322**.

(50) From the above, it is to be appreciated that defined herein is a storage compartment assembly including a storage compartment, a plurality of artificial muscles disposed in the storage compartment, and a plurality of artificial muscle cantilevers extending across an opening of the storage compartment. The artificial muscles are configured to actuate to expand in the storage compartment to contact and maintain an object disposed within the storage compartment. The artificial muscle cantilevers are configured to actuate to stiffen across the opening to prevent ingress and egress through the opening.

(51) While particular embodiments have been illustrated and described herein, it should be understood that various other changes and modifications may be made without departing from the spirit and scope of the claimed subject matter. Moreover, although various aspects of the claimed subject matter have been described herein, such aspects need not be utilized in combination. It is therefore intended that the appended claims cover all such changes and modifications that are within the scope of the claimed subject matter.

Claims

1. A storage compartment assembly comprising: a storage compartment comprising at least one inside surface and defining an opening for receiving an object; a proximity sensor configured to detect an object within the storage compartment; one or more artificial muscles disposed within the storage compartment along the inside surface, each artificial muscle comprising: a housing comprising an electrode region and an expandable fluid region; a dielectric fluid housed within the housing; and an electrode pair positioned in the electrode region of the housing, the electrode pair comprising a first electrode and a second electrode, the electrode pair configured to actuate between a non-actuated position and an actuated position such that actuation from the non-actuated position to the actuated position directs the dielectric fluid into the expandable fluid region to expand the expandable fluid region into the storage compartment; and a controller communicatively coupled to the proximity sensor and configured to, in response to receiving a signal from the proximity sensor indicative of the object being within the storage compartment, actuating the one or more artificial muscles to the actuated position.
2. The storage compartment assembly of claim 1, wherein the expandable fluid region of each artificial muscle is configured to contact and deform around the object positioned within the storage compartment to maintain the object within the storage compartment.
3. The storage compartment assembly of claim 1, wherein the at least one inside surface comprises an end surface and one or more sidewalls extending from the end surface, and the one or more artificial muscles comprises a plurality of artificial muscles disposed within the storage compartment along the sidewalls and the end surface.
4. The storage compartment assembly of claim 3, further comprising: a compartment portal extending over the opening, the compartment portal comprising a plurality of artificial muscle cantilevers, each artificial muscle cantilever comprising: a housing comprising a cantilevered region and an electrode region fluidly coupled to the cantilevered region, the cantilevered region having at least one expandable fluid chamber; a first electrode and a second electrode each disposed in the electrode region of the housing; and a bladder housing a fluid and extending into the cantilevered region, wherein the cantilevered region of the housing of each of the plurality of artificial muscle cantilevers extends over the opening.
5. The storage compartment assembly of claim 4, wherein the electrode region of each of the plurality of artificial muscle cantilevers is disposed at a periphery of the opening of the storage compartment.
6. The storage compartment assembly of claim 4, wherein the first electrode and the second electrode of the plurality of artificial muscle cantilevers are actuatable between a non-actuated position and an actuated position such that actuation from the non-actuated position to the actuated position directs the fluid in the bladder into the at least one expandable fluid chamber of the cantilevered region, expanding at least one expandable fluid chamber and increasing a stiffness of the cantilevered region of the housing.
7. The storage compartment assembly of claim 4, wherein an adjacent pair of the plurality of artificial muscle cantilevers is coupled together along respective cantilevered regions.
8. The storage compartment assembly of claim 4, wherein: the plurality of artificial muscle cantilevers comprises a first set of artificial muscle cantilevers having cantilevered regions extending over a first region of the opening and a second set of artificial muscle cantilevers having cantilevered regions extending over a second region of the opening; and ends of the cantilevered regions of the first set of artificial muscle cantilevers face ends of the cantilevered regions of the second set of artificial muscle cantilevers at a location over the opening.
9. The storage compartment assembly of claim 8, wherein contact between the ends of the cantilevered regions of the first set of artificial muscle cantilevers and the ends of the cantilevered

regions of the second set of artificial muscle cantilevers defines a central axis extending across the compartment portal, and each of the cantilevered regions of the first set of artificial muscle cantilevers and the cantilevered regions of the second set of artificial muscle cantilevers are angled obliquely from the central axis.

10. The storage compartment assembly of claim 1, wherein the storage compartment extends into one of a dashboard or a panel of a vehicle.

11. A storage compartment assembly comprising: a storage compartment comprising at least one inside surface and defining an opening for receiving an object; and a compartment portal extending over the opening, the compartment portal comprising a plurality of artificial muscle cantilevers, each artificial muscle cantilever comprising: a housing comprising an electrode region fluidly coupled to a cantilevered region, the cantilevered region having at least one expandable fluid chamber; a first electrode and a second electrode each disposed in the electrode region of the housing; and a bladder housing a fluid and extending into the cantilevered region, wherein the cantilevered region of the housing of each of the plurality of artificial muscle cantilevers extends over the opening.

12. The storage compartment assembly of claim 11, wherein the first electrode and the second electrode of the plurality of artificial muscle cantilevers are actuatable between a non-actuated position and an actuated position such that actuation from the non-actuated position to the actuated position directs the fluid in the bladder into the at least one expandable fluid chamber of the cantilevered region, expanding at least one expandable fluid chamber and increasing a stiffness of the cantilevered region of the housing.

13. The storage compartment assembly of claim 11, wherein the electrode region of each of the plurality of artificial muscle cantilevers is disposed at a periphery of the opening of the storage compartment.

14. The storage compartment assembly of claim 11, wherein an adjacent pair of the plurality of artificial muscle cantilevers is coupled together along respective cantilevered regions.

15. The storage compartment assembly of claim 11, wherein: the plurality of artificial muscle cantilevers comprises a first set of artificial muscle cantilevers having cantilevered regions extending over a first region of the opening and a second set of artificial muscle cantilevers having cantilevered regions extending over a second region of the opening; and ends of the cantilevered regions of the first set of artificial muscle cantilevers face ends of the cantilevered regions of the second set of artificial muscle cantilevers at a location over the opening.

16. The storage compartment assembly of claim 15, wherein contact between the ends of the cantilevered regions of the first set of artificial muscle cantilevers and the ends of the cantilevered regions of the second set of artificial muscle cantilevers defines a central axis extending across the compartment portal, and each of the cantilevered regions of the first set of artificial muscle cantilevers and the cantilevered regions of the second set of artificial muscle cantilevers are angled obliquely from the central axis.

17. The storage compartment assembly of claim 11, further comprising: a proximity sensor configured to detect an object within a predetermined proximity of the compartment portal; and a controller communicatively coupled to the proximity sensor and configured to, in response to receiving a signal from the proximity sensor indicative of the object being within the predetermined proximity of the compartment portal, actuating the plurality of artificial muscle cantilevers to a non-actuated position.

18. A method of operating a storage compartment assembly, the method comprising: actuating a plurality of artificial muscle cantilevers of a compartment portal, each artificial muscle cantilever comprising: a housing comprising an electrode region fluidly coupled to a cantilevered region, the cantilevered region having at least one expandable fluid chamber; a first electrode and a second electrode each disposed in the electrode region of the housing; and a bladder housing a fluid and extending into the cantilevered region, wherein the cantilevered region of the housing of each of

the plurality of artificial muscle cantilevers extends over an opening for receiving an object, the opening defined by a storage compartment, wherein the first electrode and the second electrode of the plurality of artificial muscle cantilevers are actuatable between a non-actuated position and an actuated position such that actuation from the non-actuated position to the actuated position directs the fluid in the bladder into the at least one expandable fluid chamber of the cantilevered region; and actuating one or more artificial muscles disposed within the storage compartment, each artificial muscle comprising: a housing comprising an electrode region and an expandable fluid region; a dielectric fluid housed within the housing; and an electrode pair positioned in the electrode region of the housing, the electrode pair comprising a first electrode and a second electrode, the electrode pair configured to actuate between a non-actuated position and an actuated position such that actuation from the non-actuated position to the actuated position directs the dielectric fluid into the expandable fluid region.

19. The method of claim 18, further comprising: detecting an object within a predetermined proximity of the compartment portal; and actuating the plurality of artificial muscle cantilevers from the actuated position to the non-actuated position in response to detecting the object within the predetermined proximity of the compartment portal.

20. The method of claim 19, wherein: actuating the plurality of artificial muscle cantilevers comprises actuating the first electrode and the second electrode of the artificial muscle cantilevers from the non-actuated position to the actuated position.